Vertical AI Agents

From coding to tax

AI matured first in software. The same playbook is coming to tax — here's the map, what we're building at Meta, and what we got wrong along the way.

Less a roadmap than a year of learning.

J Gu · PM, Tax & Finance @ Meta

WHERE WE'RE GOING

The arc — four quick acts, then your pushback

01	**The pattern**	Coding is the leading indicator — AI evolves in three stages
02	**Our bet**	One AI foundation across the whole tax lifecycle
03	**The reality**	40+ skills in a year — and what humbled us
04	**The how**	The infrastructure an AI agent actually needs

And throughout — **argue with me.** Most of this is still a work in progress.

WHERE WE STAND

The same three stages — coding leads, tax follows

AI's first act was chat; the next is **coding** — real work, not just talk. Coding is our leading indicator. Here's coding vs **Meta's tax build** vs the broader tax field (0–100).

STAGE 1

AI-Native Foundation

AI at every surface of the workflow — not a bolt-on.

Coding	90
Meta	80
Field	15

CODING API · Chat · IDE · review

STAGE 2

Continuous Learning

Learns from edits & outcomes; accuracy compounds.

Coding	40
Meta	10
Field	5

HARD only coding truly trains — rest is skills + context

STAGE 3

Proactive Management

Acts before you ask — a human still approves.

Coding	20
Meta	5
Field	0

CODING agents open a PR · you approve

We think we're ~80% there on the foundation — probably wrong on the exact number. Stage 2 is the frontier for everyone. Meta = projected year-end · Field = most tax teams today.

WHY THIS MATTERS FOR TAX

Coding figures it out first — then it diffuses

THE FRONTIER

Coding

Most investment, most talent — the playbook gets invented here first.

DIFFUSES →

DOMAIN BY DOMAIN

- → **Tax** startups now starting to build here
- + **Legal** contracts, research, review
- + **Healthcare** notes, coding, claims
- + **Finance & more** each its own vertical

Each domain is its own journey. Tax is one — and the first **tax-AI startups are already appearing.** The opportunity is real; the foundation is the work.

THE GOAL — AND THE FUTURE

Tax professionals are about to live what engineers already are

Coding agents didn't replace software engineers — they **amplified** them. AI will do the same for tax.

SOFTWARE ENGINEER · TODAY

🧑‍💻 + coding agent

- Agents write the boilerplate & repetitive code
- The engineer directs, reviews & decides
- Freed for **architecture & judgment** — ships far more

SAME STORY

TAX PROFESSIONAL · NEXT

🧑‍💼 + tax AI

- AI does the assembly across systems & the repetitive work we outsource today
- The professional directs, reviews & signs off
- Freed for **planning, strategy & advising** — covers far more

No one's replaced. The work moves **up** — to judgment, strategy, and impact.

OUR PLAY AT META

So we start at Stage 1 — across the whole tax lifecycle

We're building **one** AI-native foundation — and every stage of the tax lifecycle draws from it.

Tax Planning

Scenario modeling & structuring, grounded in our positions and the law.

Compliance & Reporting

High-volume, rules-based work on real-time data — the natural first mover.

Tax Audit

Controversy & defense on the same lineage, sources, and document trail.

1 **One AI-Native Foundation** shared data · grounding · governance — under all three

Foundation first. Build it once, build it well.

WHERE WE ARE — 12 MONTHS IN

From a standing start to 40+ skills in a year

Q2 ’25	**12 mo**	**40+**	**3**
foundation journey began	in, and compounding	skills shipped	lifecycle areas live

Planning

Skills for **new product initiatives** — model the tax impact of a launch before it ships.

Compliance & Reporting

In testing now — **live by year-end** with very high coverage across **10+ countries**.

Audit — an end-to-end skill

1. One-click upload
2. Extract audit-critical info
3. Auto-search every DB & internal system
4. Draft the audit response
5. Reference historical audits

Real skills, in production — and each one makes the foundation stronger.

Three things that humbled us

1 · “Just ask the AI to do it”

WE ASSUMED

Hand it the audit — it writes the response.

WE LEARNED

AI is an intern. You teach it where the data lives, the reply format, your risk criteria — and document all of it. It's a long learning curve.

2 · “What's an Eval?” — nobody knew, 9 months ago

WE ASSUMED

Past audit responses are our answer key.

WE LEARNED

Last year's answer isn't valid this year — **even for the same question.** Evals have to be built on **today's fact patterns.**

3 · Context is not data

WE ASSUMED

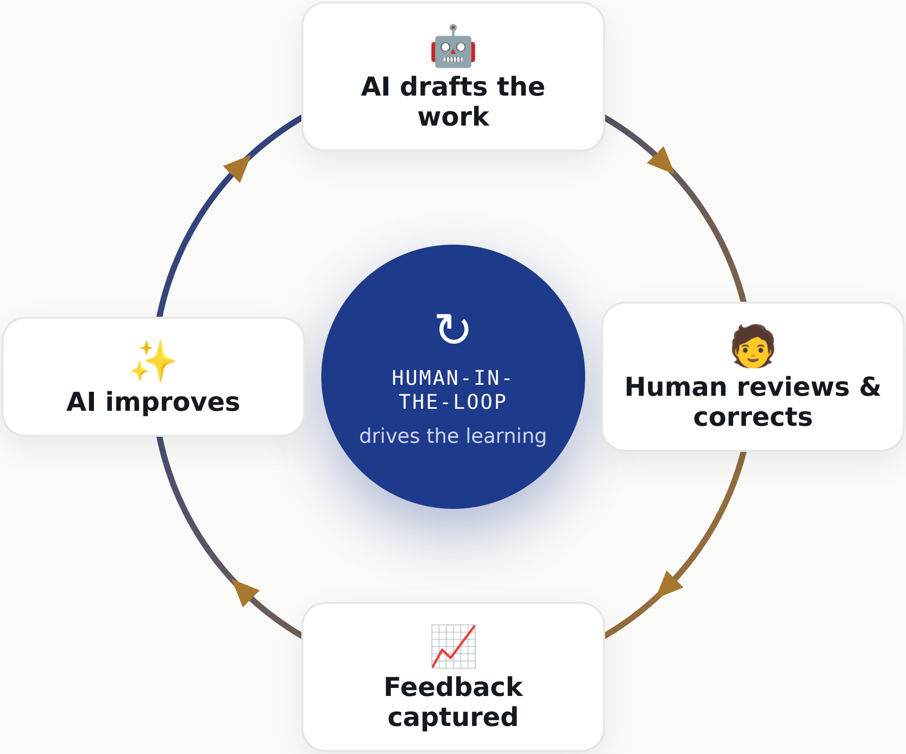
Drop it all in Google Drive — even a spreadsheet — and call it knowledge.

WE LEARNED

Drive is fine for **context** (unstructured knowledge). But **data belongs in a database**, not a doc — that's why the foundation & graph matter.

AI isn't magic. It's an intern you have to train — and **we're still learning how.**

Humans in the loop — the engine behind continuous learning



No true continuous learning — yet

- Today we turn the wheel **by hand** — people correct, the AI improves via **skills + context**.
- Only the **coding** vertical actually **trains the model** in the loop.
- For tax — and every other vertical — real continuous learning is **unsolved, and hard to scale**.

Human-in-the-loop is how we improve now — and the **data engine** for continuous learning when it arrives.

THE MENTAL MODEL THAT UNLOCKED IT

Treat AI like a brilliant intern

Fast, capable, eager — and on day one it knows **nothing about your world**. You don't just hand it the work. You onboard it.

📍
Where the data lives
Which systems, which fields, what's the source of truth.

📝
What "good" looks like
The reply format, the structure, the tone of a real answer.

⚖️
Your judgment criteria
How you weigh risk — the calls only experience makes.

...and you **write it all down.** The teaching *is* the work — and the work is infrastructure.

WHAT THE INTERN NEEDS

Onboard the intern — and each thing it needs is infrastructure

Same as a new hire: give it a brain, skills, hands, memory, and your knowledge — **each one is a piece of infrastructure.**

THE AGENT NEEDS…		…WHICH IS INFRASTRUCTURE
🧠 **A brain** — to reason & decide	→	Foundational AI model
🧩 **Skills** — the tasks it can do	→	Skill library + management
🖐️ **Hands** — to act in your systems	→	Tool calling (MCP)
💾 **Memory** — to remember past work	→	Memory store
📚 **Knowledge** — your tax law & data	→	Data foundation
🕸️ **Relationships** — how entities, jurisdictions & products connect	→	Knowledge graph
🎯 **Context** — the task in front of it	→	Grounding / retrieval
🛡️ **Guardrails** — judgment & sign-off	→	Trust & controls

The model is just the brain. **Everything else is the infrastructure** — and that's the work.

MAKE IT CONCRETE

Why the model alone isn't enough

“Did we charge the right sales tax on this $2M order shipped to Texas?”

The model alone

Knows Texas sales tax *in general*. But not **your** order, your nexus, today's rate — or how to prove it.

With the foundation, it can…

Data platform	pull the real order — $2M, ship-to TX, the product SKU
Graph	which legal entity sold it · TX state→county→city · is the product taxable
Knowledge	today's Texas rule & rate — **with a citation**
Models	reason over all of it to an answer
Trust & controls	check who may see it · log every step
Eval + lineage	confidence check; unsure → human · every number traces back

The model is one slice of the answer. The **infrastructure is the rest** — and the part that holds up in an audit.

THE HONEST PART

Genuinely hard — we just had a head start

Why it's hard

- Fragmented, messy data across ERPs & jurisdictions
- Real-time / streaming is a specialized skill set
- Talent fluent in tax *and* AI is doubly rare
- Split ownership: Tax vs. Finance Systems vs. IT
- Tax is a cost center — hard to fund infra

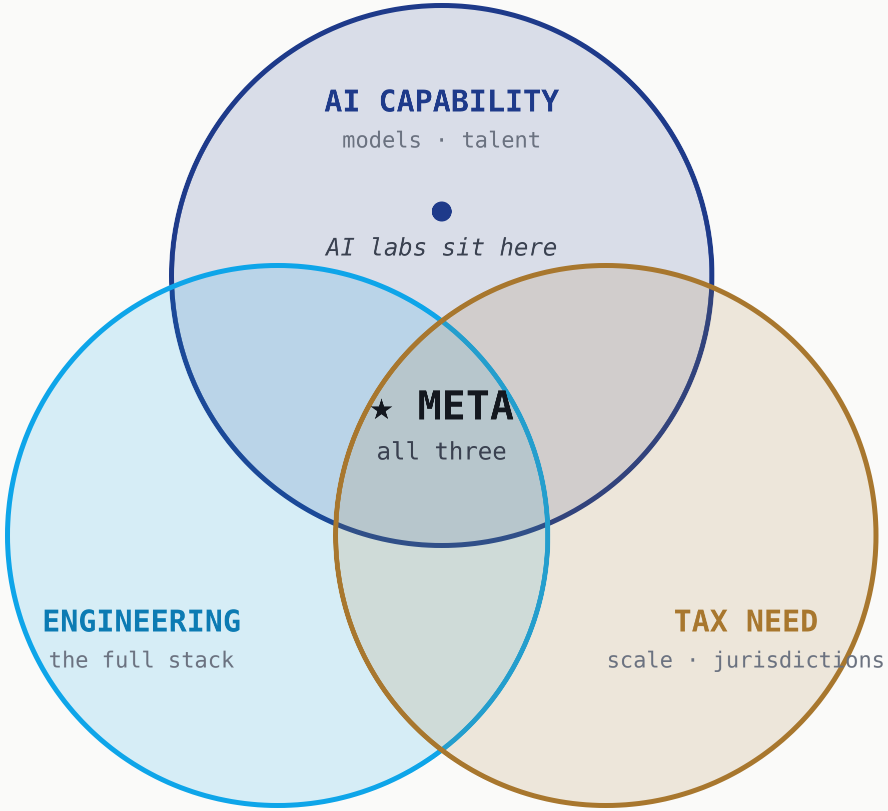
Why we had a head start

- Hyperscale data infrastructure already exists
- Real-time / event-driven by default
- World-class platform, data & ML engineers
- AI platform & compute in-house, not procured
- Engineering-owned tax tech — ships like a product team

The barriers most teams hit were largely **already solved here** — so we got a running start. That's luck, not genius.

WHY WE GOT A RUNNING START

The transformation takes all three at once



The frontier labs? Not yet.

- They're building the **ecosystem** — models, skills, MCP — and want you on their platform.
- But they have **many verticals to chase** — tax is just one of them.
- So they'll ship **generic skills** — but won't go deep into *your* process and data.

The horizontal AI is theirs to sell. The **vertical — your process, your data — is yours to own.** We just happened to already have all three.

THE TAKEAWAY

Tax is where coding was a few years ago.

We don't have to predict the future — we get to read it. Build the foundation now, treat the AI like the intern it is, and let your people move up to judgment.

Foundation first | Teach the intern | People move up

I'm still figuring this out. Where do you think I'm wrong?

Thank you · J Gu · PM, Tax & Finance @ Meta